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to the diffuse aurora but are not responsible for the pulsating patches (26).

Theoretical calculations (9, 27) using the observed lower-band chorus amplitudes (~50 pT) show that such amplitudes are sufficiently strong to cause precipitation of ~10-keV electrons into the atmosphere. Wave burst observations with higher resolution (fig. S1) indicate more intense fields (~30 mV/m and ~1 nT) that were quasi-field-aligned and propagated away from the equator, making the interaction with electrons that precipitate into the atmosphere and drive PA substantially more efficient near the equator (less than ~15°) because of Landau damping and increasing resonant energy in latitude. Each chorus element generated near the equator with a characteristic size of ~100 to 3000 km (7, 28) and the oblique propagation away from the original magnetic field line (10, 29) would lead to interaction regions with a typical patch size (~100 km in the ionosphere and ~a few thousand km in the magnetosphere).

Magnetic field line mapping has been a problematic issue in magnetosphere-ionosphere coupling studies. Based on the property that the high-correlation region occurs over a single auroral patch with the correlation coefficient outside this area diminishing quickly with distance, PA can be used to highlight the physical link between chorus wave activity and each auroral pul-

sation. The PA thus provides a unique opportunity to identify the footprint of the magnetic field line threading the spacecraft, to a precision within the auroral patch size (less than ~100 km) and possibly even down to a few imager pixels (~km).

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24. We also analyzed other events (table S1).
25. Weak ECH waves that may lead to weak auroral emission were also observed in the electric field spectra above 1.0 f<sub>c</sub>.
26. The auroral intensity did not diminish following the decrease in the ECH intensity but stayed at a high level. This is attributable to the long lifetime (~1 min) of excited oxygen atom at higher altitudes (32) due to the precipitation of lower-energy electrons (~1 keV), as expected from resonance with ECH (11).
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# Supporting Online Material

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Fig. S1

Table S1

Movies S1 and S2

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## Cellodextrin Transport in Yeast for Improved Biofuel Production

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Fungal degradation of plant biomass may provide insights for improving cellulosic biofuel production. We show that the model cellulolytic fungus *Neurospora crassa* relies on a high-affinity cellodextrin transport system for rapid growth on cellulose. Reconstitution of the *N. crassa* cellodextrin transport system in *Saccharomyces cerevisiae* promotes efficient growth of this yeast on cellodextrins. In simultaneous saccharification and fermentation experiments, the engineered yeast strains more rapidly convert cellulose to ethanol when compared with yeast lacking this system.

The bioethanol industry uses the yeast *Saccharomyces cerevisiae* to ferment sugars derived from cornstarch or sugarcane into ethanol (1). Plant cell walls provide an abundant alternate sugar source yet remain largely unused (2, 3). Present strategies for using plant biomass use large quantities of cellulase cocktails to release

glucose from plant cell walls, posing economic and logistical challenges for implementation of these processes (4). Furthermore, *S. cerevisiae* cannot ferment the cellodextrins naturally released by cellulases (5, 6) and require cellulase cocktails supplemented by  $\beta$ -glucosidases to quantitatively produce fermentable glucose (7).

In contrast to *S. cerevisiae*, cellulolytic fungi grow well on cellodextrins. The model cellulolytic fungus *Neurospora crassa*, when grown on pure cellulose, increases transcription of seven major facilitator superfamily (MFS) sugar transporters as well as an intracellular  $\beta$ -glucosidase (8). A strain carrying a deletion for one of these predicted transporters, NCU08114, grew slowly on cellulose (fig. S1) (9), and strains lacking either NCU08114 or NCU00801 poorly consumed cellobiose, cellotriose, and cellotetraose (fig. S2).

Orthologs of NCU08114 and NCU00801 are widely distributed in the fungal kingdom (Fig. 1A) (10–13), and recent whole-genome transcriptional profiling studies show their importance to interactions between fungi and plants. For example, some cellulolytic fungi increase expression levels of NCU08114 orthologs while degrading plant wall material (10), and the Périgord black truffle increases the expression of a NCU00801 ortholog during symbiotic interactions with plant roots (11). In addition, certain yeasts that grow on cellobiose contain orthologs of NCU08114 and NCU00801 (12). All of these organisms also contain genes for intracellular  $\beta$ -glucosidases (10), suggesting that cellodextrin transport systems are widespread in nature and are essential for optimal growth of fungi on cellulose-derived sugars.

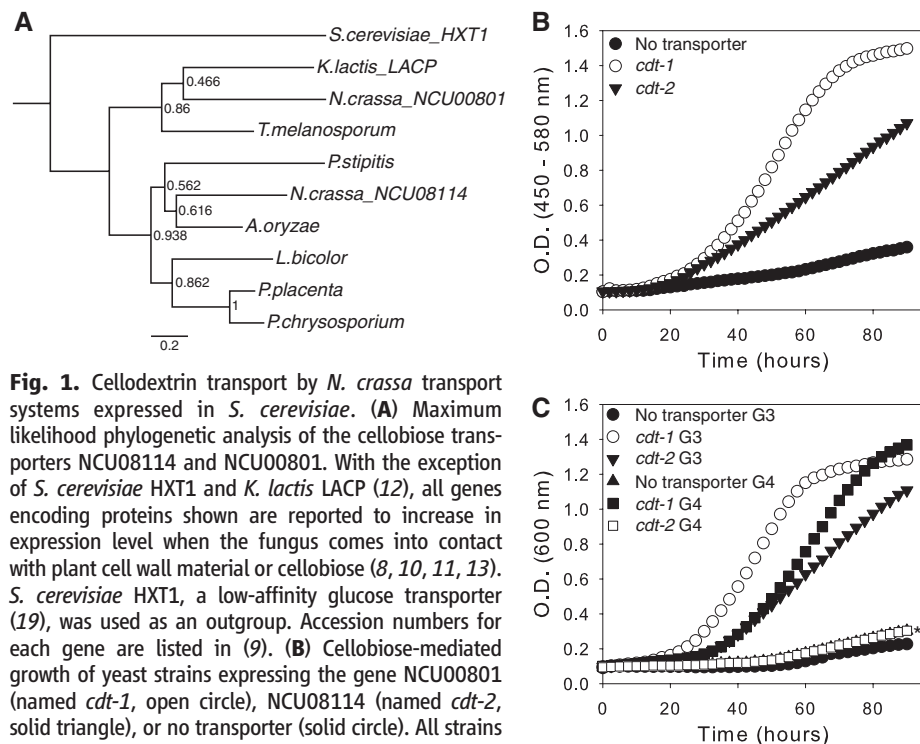
Because cellobiose is not catabolized by *S. cerevisiae* (5, 6) and is not accumulated in the cytoplasm (fig. S3), we reasoned that expression of a functional cellodextrin transport system from *N. crassa* might allow *S. cerevisiae* to grow with cellobiose as the sole carbon source. Yeast strains expressing NCU00801 or NCU08114, together with the intracellular  $\beta$ -glucosidase NCU00130 (hereafter named GH1-1), grew on cellobiose at rates of about 30 and 12% of the rate of *S. cerevisiae* on glucose, respectively (Fig. 1B and fig. S4). Cellodextrins longer than cellobiose also support the growth of yeast expressing these transporters (Fig. 1C), indicating that cellodextrins longer than cellobiose are transported by NCU00801 and NCU08114, hereafter called CDT-1 and CDT-2,

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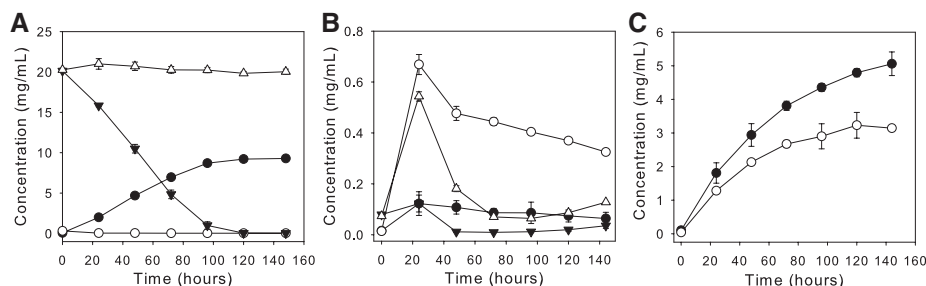
\*To whom correspondence should be addressed. E-mail: jcate@lbl.gov

respectively. Growth cannot be explained by the extracellular hydrolysis of cellodextrins to glucose followed by transport because a strain expressing only the intracellular  $\beta$ -glucosidase GH1-1 grew slowly on cellodextrins (Fig. 1, B and C, and fig. S4). Fur-

thermore, the activity of  $\beta$ -glucosidase GH1-1—which is able to hydrolyze cellobiose, cellotriose, and cellotetraose (fig. S5)—is negligible in culture supernatants and cannot explain the growth of yeast expressing the transporters (fig. S6).



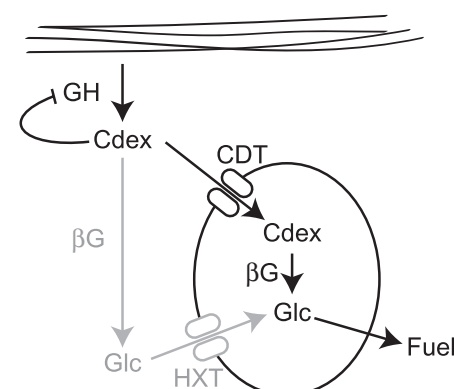
**Fig. 1. Celldextrin transport by *N. crassa* transport systems expressed in *S. cerevisiae*.** (A) Maximum likelihood phylogenetic analysis of the cellobiose transporters NCU08114 and NCU00801. With the exception of *S. cerevisiae* HXT1 and *K. lactis* LACP (12), all genes encoding proteins shown are reported to increase in expression level when the fungus comes into contact with plant cell wall material or cellobiose (8, 10, 11, 13). *S. cerevisiae* HXT1, a low-affinity glucose transporter (19), was used as an outgroup. Accession numbers for each gene are listed in (9). (B) Cellobiose-mediated growth of yeast strains expressing the gene NCU00801 (named *cdt-1*, open circle), NCU08114 (named *cdt-2*, solid triangle), or no transporter (solid circle). All strains also express the intracellular  $\beta$ -glucosidase, NCU00130 (named *gh1-1*). A representative experiment is shown. Growth rates from three independent experiments are as follows: *cdt-1*,  $0.0341 \pm 0.0010$  hours<sup>-1</sup>; *cdt-2*,  $0.0131 \pm 0.0008$  hours<sup>-1</sup>; and no transporter,  $0.0026 \pm 0.0001$  hours<sup>-1</sup> (mean  $\pm$  SD). (C) Growth of yeast strains on cellotriose and cellotetraose. Strains expressing the intracellular  $\beta$ -glucosidase, *gh1-1*, as well as the transporters listed in the legend, were grown with 0.5% (w/v) of cellotriose (G3) or cellotetraose (G4) serving as the sole carbon source. A representative experiment is shown. Growth rates from three independent experiments are as follows: *cdt-1* cellotriose,  $0.0332 \pm 0.0004$  hours<sup>-1</sup>; *cdt-1* cellotetraose,  $0.0263 \pm 0.0020$  hours<sup>-1</sup>; no transporter cellotriose,  $0.0043 \pm 0.0015$  hours<sup>-1</sup>; *cdt-2* cellotriose,  $0.0178 \pm 0.0005$  hours<sup>-1</sup>; *cdt-2* cellotetraose,  $0.0041 \pm 0.0003$  hours<sup>-1</sup>; and no transporter cellotetraose,  $0.0031 \pm 0.0008$  hours<sup>-1</sup> (mean  $\pm$  SD). The data for yeast growth on cellotetraose with *cdt-2* overlaps the data in the absence of transporter (asterisk).



**Fig. 2. Cellobiose fermentation, and simultaneous saccharification and fermentation of cellulose, by *S. cerevisiae* expressing the cellobiose transport system from *N. crassa*.** (A) Cellobiose fermentation to ethanol. Ethanol was produced by yeast strains with CDT-1 (solid circle) or without CDT-1 (open circle) under anaerobic conditions. Also shown are cellobiose concentrations during the fermentation reaction using yeast strains with CDT-1 (solid triangle), or without CDT-1 (open triangle). (B) SSF using yeast strains with and without CDT-1, under anaerobic conditions. Shown are cellobiose (solid circle) and glucose (solid triangle) concentrations in the presence of a strain with CDT-1, and cellobiose (open circle) and glucose (open triangle) concentrations in the presence of a strain lacking CDT-1. 0.1 mg/ml cellobiose = 292  $\mu$ M. (C) Ethanol produced during SSF by using a strain with CDT-1 (solid circle) or without CDT-1 (open circle). In (A) to (C), values are the mean of three biological replicates. Error bars are the SD between these replicates. All strains also express the intracellular  $\beta$ -glucosidase, *gh1-1*.

To directly assay transporter function, the uptake of [<sup>3</sup>H]-cellobiose into yeast cells was measured. Both CDT-1 and CDT-2 are high-affinity cellobiose transporters, with Michaelis constant ( $K_m$ ) values of  $4.0 \pm 0.3$   $\mu$ M and  $3.2 \pm 0.2$   $\mu$ M (mean  $\pm$  SD), respectively (fig. S7). The expression-normalized maximum velocity ( $V_{max}$ ) of CDT-1 is over twice that of CDT-2 (fig. S7), which might explain differences in the observed growth rates of the yeast strains (Fig. 1, B and C). Cellobiose transport by CDT-1 and CDT-2 is inhibited by excess cellobiose, and CDT-1 activity is also inhibited by cellotetraose (fig. S8). These data further support that CDT-1 and CDT-2 directly transport the longer celldextrins in growth assays (Fig. 1C). Thus, the combinations of CDT-1 or CDT-2 with GH1-1 constitute fully functional celldextrin transport systems.

We next tested whether, in addition to growth, a complete celldextrin catabolism pathway would be useful in *S. cerevisiae* for lignocellulosic bio-fuel production. Fungal cellulases evolved to function in conjunction with the consumption of sugars released from plant cell walls and work most effectively in this context (14). Simultaneous saccharification and fermentation (SSF) mimics this natural synergy by the addition of fermenting microbes to the plant biomass depolymerization reaction, resulting in a more efficient process (15, 16). Current SSF schemes are limited by the need for quantitative conversion of cellulose to glucose outside yeast cells (7). A celldextrin transport system could be particularly useful during SSF because it would remove the requirement for full hydrolysis of cellulose to glucose by extracellular  $\beta$ -glucosidases (17) and would reduce the risk of contamination by glucose-dependent organisms (18). Furthermore, both CDT-1 and CDT-2 have a higher apparent affinity for cellobiose ( $K_m \approx 3$  to 4  $\mu$ M) (fig. S7) when compared with secreted fungal  $\beta$ -glucosidases



**Fig. 3. Use of celldextrin transport pathways from filamentous fungi in yeast during simultaneous saccharification and fermentation of cellulose.** The celldextrin (Cdex) transport pathway (black) includes a celldextrin transporter (CDT) and intracellular  $\beta$ -glucosidase ( $\beta$ G). The sugar catabolism pathway present in standard yeast includes hexose transporters (HXT). In SSF, cellulases (GH) and extracellular  $\beta$ -glucosidase ( $\beta$ G) may both be used.

[ $K_m \approx 100$  to  $1000 \mu\text{M}$  (17)] and to *S. cerevisiae* hexose transporters' apparent affinity for glucose [ $K_m \approx 1000$  to  $10,000 \mu\text{M}$  (19)]. Therefore, the cellodextrin transport systems should more effectively maintain soluble sugar levels below the concentration at which they inhibit fungal cellulases [inhibition constant ( $K_i$ ) of cellobiose  $\approx 19$  to  $410 \mu\text{M}$  (20)].

With little optimization, yeast expressing *cdt-1* and *ghl-1* fermented cellobiose with an ethanol yield of  $0.441 \pm 0.001$  (grams of ethanol/grams of glucose  $\pm$  SD), which is 86.3% of the theoretical value (Fig. 2A) (21). This yield is close to present industrial yields of ethanol from glucose of 90 to 93% (22). Yeasts expressing a cellodextrin transport system markedly improve the efficiency of SSF reactions by reducing the steady-state concentration of both cellobiose and glucose and by increasing the ethanol production rate (Fig. 2, B and C). The addition of a cellodextrin transport system to biofuel-producing strains of yeast (Fig. 3) overcomes a major bottleneck to fermentation of lignocellulosic feedstocks and probably will help to make cellulosic biofuels economically viable.

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## Supporting Online Material

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Materials and Methods

Figs. S1 to S8

References

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# Sequencing of *Culex quinquefasciatus* Establishes a Platform for Mosquito Comparative Genomics

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*Culex quinquefasciatus* (the southern house mosquito) is an important mosquito vector of viruses such as West Nile virus and St. Louis encephalitis virus, as well as of nematodes that cause lymphatic filariasis. *C. quinquefasciatus* is one species within the *Culex pipiens* species complex and can be found throughout tropical and temperate climates of the world. The ability of *C. quinquefasciatus* to take blood meals from birds, livestock, and humans contributes to its ability to vector pathogens between species. Here, we describe the genomic sequence of *C. quinquefasciatus*: Its repertoire of 18,883 protein-coding genes is 22% larger than that of *Aedes aegypti* and 52% larger than that of *Anopheles gambiae* with multiple gene-family expansions, including olfactory and gustatory receptors, salivary gland genes, and genes associated with xenobiotic detoxification.

Mosquitoes are the most important vectors of human disease and are responsible for the transmission of pathogens

that cause malaria (*Anopheles*), yellow fever and dengue (*Aedes*), as well as lymphatic filariasis and encephalitis viruses (*Culex*, *Aedes*, *Anopheles*).

Sequencing the *Anopheles gambiae* and *Aedes aegypti* genomes has provided important insights into the genomic diversity underlying the complexity of mosquito biology (1, 2). We describe the sequencing of the *Culex quinquefasciatus* (the southern house mosquito) genome, which offers a reference genome from the third major taxonomic group of disease-vector mosquitoes. With more than 1200 described species, *Culex* is the most diverse and geographically widespread of these three mosquito genera. Apart from contributing to the spread of West Nile encephalitis, it also transmits St. Louis encephalitis and other viral diseases and is a major vector of the parasitic *Wuchereria bancrofti* nematode that has caused the majority of the 120 million current cases of lymphatic filariasis (3).

Taxonomy of the *Culex pipiens* species complex is the subject of a long-standing debate, an issue complicated by the occurrence of viable species hybrids in many geographic areas [reviewed in (4, 5)]. We followed the standard set by the National Center for Biotechnology Information and refer to the species sequenced here as *C. quinquefasciatus*. The Johannesburg strain of *C. quinquefasciatus* was established from a single pond in Johannesburg, South Africa—an area where the two taxa, *C. quinquefasciatus* and *C. pipiens*, were found to be sympatric [(5) therein described as subspecies *C. pipiens quinquefasciatus* and *C. pipiens pipiens*] but have remained much more genetically distinct than the same two sympatric taxa found in California.

We were able to map 9% of the *C. quinquefasciatus* genes (1768 genes) on the three chromosomes with the use of published and new *C. quinquefasciatus* and *Ae. aegypti* markers (6). Of these mapped genes, 803 had *An. gambiae* orthologs and 641 had *Drosophila melanogaster*